

- Q.26 Distinguish between AC Bridges and DC bridges.
- Q.27 Explain the Schering bridge circuit with the help of diagram.
- Q.28 Briefly discuss about digital storage oscilloscope.
- Q.29 Explain pulse generator.
- Q.30 Write short note on PMMC.
- Q.31 Explain different types of errors that occur in measurements.
- Q.32 Explain the process of calibration.
- Q.33 Write importance and applications of standards and calibration.
- Q.34 Write limitations of digital multi-meters.
- Q.35 Explain block diagram of a digital multi-meter.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Draw the block diagrams of a general -purpose oscilloscope and explain function of each block in detail.
- Q.37 Explain working, need and characteristics of instrument amplifier.
- Q.38 Explain the following
- DC voltage and DC Current
 - AC voltage and AC current

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3rd Sem / Electronics and Communication Engineering EI, Med. Eltx, Mechagronics (6th Sem), GE (4th Sem), Power Eltx, Elect & Eltx. Engg.

Sub : Electronic instruments and measurement

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The waveform is stored in _____.
 a) Compressed form b) Analog form
 c) Digital form d) Mixed form
- Q.2 The analog signal is digitized using _____.
 a) D/A converter b) Oscillator
 d) A/D converter d) Rectifier
- Q.3 A Multiplier is _____.
 a) Non Capacitive b) Capacitive
 c) Non inductive d) Resistive
- Q.4 A simple bridge circuit consists consists of a network of _____.
 a) 3 resistance arms b) 2 resistance arms
 c) 4 resistance arms d) 6 resistance arms
- Q.5 Which principle operates a bridge circuit?
 a) Null indication b) Ampere's rule
 c) Partial indication d) Kirchhoff's laws

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- Q.6 How is the waveforms adjusted?
- by adjusting the voltage
 - Through shift controls
 - By reducing the current
 - By means of a galvanometer
- Q.7 In a Wien-bridge oscillator for obtaining 160 Hz frequency output will be the capacitor value if resistance is selected as $1K\Omega$?
- 10nF
 - 1nF
 - 1 nF
 - 10 nF
- Q.8 Continuous voltages (or) current signals are measured using
- Tachometers
 - Sonometers
 - Analog meter
 - Digital meters
- Q.9 What is the reason for the occurrence of systematic errors in an instrument?
- No use for a long time
 - High use
 - Manufacturing fault
 - Delivery fault
- Q.10 By making use of a CRO _____.
- Many characteristics of a signal can be measured
 - Only a few characteristics of a signal can be measured
 - No characteristics of a signal can be measured
 - Signal can only be displayed

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write full form of DSO.
- Q.12 Peak to peak voltage is given by _____.
- Q.13 Write full form of PMMC.
- Q.14 Write an example of a digital electronic.
- Q.15 What type of signal does a digital instrument display?
- Q.16 What are the different types DVM?
- Q.17 An induction meter can handle current upto _____.
- Q.18 Usually how many types of errors are present in scientific measurements?
- Q.19 Digital meters are preferable than analog meters.
- Q.20 An ideal voltmeter function as _____ circuit.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write short note on CRT.
- Q.22 Describe with diagram the operation of a Logic analyzer.
- Q.23 Compare analog and digital instrument.
- Q.24 write five applications of digital millimeter.
- Q.25 Discuss the working principle of Q-meter & its applications

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